Course Title: Software Engineering (3 Cr.)

Course Code: CACS253

Year / Semester: II / IV

Class Load: 4 Hrs. / Week (Theory: 3 Hrs. Tutorial: 1)

Course Description

This course includes the topics that provide fundamental concept and standard of software engineering so that students will be able to develop software and/or handle software project using the global standard of software.

Course Objectives

The course has the following objectives:

- To provide the students with the basic competencies required to identify requirements of developed system.
- To understand and document the system design and maintain a developed system.
- To presume the general understanding of computer and computer programming.

Course Detail

Specific Objectives	Course Content	Hours	References
• Define Software. • Classify various types of software. • Explain various characteristics of software. • Explain the attributes of good software. • Describe software engineering. • Identify key challenges that software engineering are facing. • Explain system engineering and software engineering • Explain professional practice.	**Unit1 : Introduction** 1.1 Definition of Software, Types of Software, Characteristic of Software, Attributes of Good Software. 1.2 Definition of Software Engineering, Costs, Key Challenges that Software Engineering Facing 1.3 System Engineering and Software Engineering, Professional Practice.	4 Hrs.	1. Chapter 1 - Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010 2. Chapter 1 -Roger S. Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition 3. Chapter 1- Rajib Mall," Fundamental of Software Engineering",3rd Edition, 2010
• Define Software process. What is software process model. • Explain the waterfall model, evolutionary development. • Describe componentbased software engineering. • Explain process iteration, incremental delivery and software development. Describe their advantages and disadvantages. • Explain agile methods • Explain the importance of extreme programming. • Explain rapid application development. • Describe software prototyping. • Describe rational unified process. • Explain computer aided software engineering. • Classify the various CASE tools.	**Unit 2 : Software Development Process Model** 2.1 Software Process, Software Process Model: The Waterfall Model, Evolutionary Development, Component Based Software Engineering(CBSE), Process Iteration, Incremental Delivery, Spiral Development, Rapid Software Development, Agile Methods, Extreme Programming, Rapid Application Development 2.2 Software Prototyping; Rational Unified Process (RUP), Computer Aided Software Engineering (CASE); Overview of CASE Approach	8 Hrs.	1. Chapter 2 – Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010 Chapter 3- Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010 2. Chapter 2- Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition Chapter 3- Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition 3. Chapter 2- Rajib Mall," Fundamental of Software Engineering",3rd Edition, 2010 Chapter 12- Rajib Mall," Fundamental of Software Engineering",3rd Edition, 2010

	Classification of CASE tools.		
• Describe system and software requirements. • Realize the types of software requirements. • Explain the functional and nonfunctional requirements. • Explain the domain requirements. • Describe user requirements. • Identify overview of techniques. • Explain the use case ethnography. • Explain requirement validation and requirement specification. • Define feasibility.	**<u>Unit 3 : Software Requirement Analysis and Specification</u>** 3.1 System and Software Requirements, Types of Software Requirements: Functional and Non Functional Requirements, Domain Requirements, User Requirements, Elicitation and Analysis of Requirements. 3.2 Overview of Techniques, View Points, Interviewing, Scenarios, Use-Case Ethnography, Requirement Validation, Requirement Specification, Feasibility.	10 Hrs.	1. Chapter 4- Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010 2. Chapter 5- Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition 3. Chapter 4 – Rajib Mall," Fundamental of Software Engineering",3rd Edition, 2010
• Define software design. • Explain the design concept. • Describe architectural design. • Explain repository model,client server model,layered model • Explain modular decomposition. • Explain procedural design using structural methods. • Explain user interface design. • Describe human computer interaction. • Describe information presentation. • Explain interface evaluation. • Describe design notation.	**<u>Unit 4 : Software Design</u>** 4.1 Design Concept; Abstraction, Architecture, Patterns, Modularity Cohesion, Coupling Information Hiding ,Functional Independence,Refinement 4.2 Architectural Design Repository Model Client Server Model , Layered Model, Modular Decomposition; 4.3 Procedural Design Using Structural Methods, User Interface Design, Human Computer Interaction, ,Information Presentation , ,Interface Evaluation ,Design Notation	10 Hrs.	1. Chapter 6- Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010 Chapter 7- Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010 2. Chapter 8- Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition Chapter 9- Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition Chapter 10- Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition Chapter 11-Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition 3. Chapter 5 – Rajib Mall," Fundamental of Software Engineering",3rd Edition, 2010 Chapter 9- Rajib Mall," Fundamental of Software Engineering",3rd Edition, 2010
• Define coding.	**<u>Unit 5 : Coding</u>**	2Hrs	1. Chapter 10- Rajib Mall,"

used for software development. • Explain various development tools • Describe good programing practices.	and Development Tools 5.2 Selecting Languages and Tools Good Programming Practices		Engineering",3rd Edition, 2010
• Explain software testing. • Differentiate verification and validation. • Explain various techniques of testing. • Explain various level of testing. • Design test cases. • Explain quality management activities. • Define product and its process. • Explain quality standards. • Describe capability maturity model.	**Unit 6 : Software Testing and Quality Assurance** 6.1. Verification and Validation , Techniques of Testing; Black-box and White-box Testing, 6.2. Inspections 6.3. Level of Testing ; Unit Testing, Regression Testing; 6.4. Design of Test Cases, Quality Management Activities, Product and Process ,Quality Management Activities , Product and Process, Quality Standards, ISO9000 , Capability Maturity Model(CMM);	6Hrs	1. Chapter 8- Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010 Chapter 26- Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010 2. Chapter 14- Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition Chapter 16- Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition Chapter 17- Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition Chapter 25-Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition
• Explain evolution nature of software • Describe different types of maintenance. • Explain functionality addition or modification. • Describe re-engineering. • Define configuration management. • Explain importance of configuration management.	**Unit 7 : Software Maintenance** 7.1. Evolving Nature of Software, Different Types of Maintenance ; Fault Repair, Software Adaptation 7.2. Functionality Addition or Modification; Maintenance Prediction, Re-Engineering, Configuration Management (CM), Importance of CM, Configuration Items, Versioning;	3Hrs.	1. Chapter 9- Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010 Chapter 25- Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010 2. Chapter 22-Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition Chapter 29-Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition 3. Chapter 10- Rajib Mall," Fundamental of Software

			Chapter 11-Rajib Mall," Fundamental of Software Engineering",3rd Edition, 2010
• Explain the need for the project management in software project. • Explain management activities. • How project is managed, planned and cost estimated. • Explain need of project scheduling. • Explain risk management.	**Unit 8: Management Software Projects** 8.1. Needs for the Proper Management of Software Projects, Management Activities; Project Planning Estimating Costs, Project Scheduling, Risk Management, Managing People.	2Hrs	1. Chapter 22- Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010 Chapter23- Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010 2. Chapter 24- Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition Chapter 25-Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition Chapter 26-Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition Chapter 27-Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition Chapter 28-Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition 3. Chapter 3-Rajib Mall," Fundamental of Software Engineering",3rd Edition, 2010 Chapter 13-Rajib Mall," Fundamental of Software Engineering",3rd Edition, 2010

Teaching Methods

The general teaching methods includes class lectures, group discussions, case studies, guest lectures, research work, project work, assignments, and exams, depending upon the nature of the topics. The teaching faculty will determine the choice of teaching pedagogy as per the need of the topics.

Evaluation

Evaluation Scheme				
Internal Assessment		External Assessment		Total
Theory	Practical	Theory	Practical	
40	-	60 (3 Hrs.)	-	100

Internal/Practical Assessment Format [FM = 40]

Internal Assessment Format [FM = 40] – Subject Teacher				
Term Examination		Assignment	Attendance	Total
Mid – Term	Pre - Final			
10	10	10	10	40

Note: Assignment may be subject specific case study, seminar paper preparation, report writing, project work, research work, presentation, problem solving etc.

Final Examination Questions Format [FM = 60, PM = 24, Time = 3 Hrs.]

SN	Question Type	Number of Questions Given	Marks per Question	Total Marks
1	Group – 'A' Objective Type Questions(Multiple Choice Questions)	10	1	10 x 1 = 10
2	Group – 'B' Short Questions (Attempt any SIX questions)	7	5	6 x 5 = 30
3	Group – 'C' Long Questions (Attempt any TWO questions)	3	10	2 x 10 = 20

- Student must pass 'Internal Assessment', 'Practical Assessment' and 'Final Examination' separately.
- Student must attend each and every activity of 'Internal Assessment' otherwise he/she will be declared as 'Not Qualified' for final Examination.

Text Books

1 Roger S.Pressman, "Software Engineering: A Practitioner 's approach", 6th Edition , McGraw Hill International edition,2005

Reference Books

1 Ali Behforooz , and Frederick J.Hudson, "Fundamental of Software Engineering", OUP,1996
2 Ian Sommerville," Software Engineering", 9th Edition Addison-Wesley, 2010.
3 Rajib Mall," Fundamental of Software Engineering",3rd Edition, 2010.
4 Pankaj Jalote, "An Integrated Approach to Software Engineering ", 2nd Edition, Springer, 1997.

Internal Assessment Marks Submission Format

Campus Name:									
Subject Name: Software Engineering						Subject Code: CACS253			
SN	TU Registration No.	Name	Symbol No.	Mid – Term [10]	Pre – Final [10]	Assignment [10]	Attendance [10]	Total [40]	Remarks

Name of Subject Teacher:

Name of Director/HoD/Coordinator:

Signature:

Signature:

Date:

Date: